

PENGYUAN WANG

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EDUCATION

Zhejiang University (ZJU), Hangzhou, China
B.Eng. in Robotics Engineering (Honors Program)

2023–2027 (expected)
GPA: 4.04/4.3 (89.09/100)

RESEARCH EXPERIENCE

Tsinghua University (THU), Beijing, China
Research Intern at THU Spatial Intelligence Lab (Supervised by Prof. [Yiming Li](#))

Oct 2025–Present

RESEARCH INTERESTS

My current research interests focus on machine learning-based control for dynamic systems, including reinforcement learning, imitation learning, perception–action loop, and sim-to-real transfer. In the long term, I hope to fully leverage the configuration and hardware advantages of diverse robotic platforms to enable safe, autonomous, and widely deployed robots in both industrial and daily-life settings.

RESEARCH PROJECTS

End-to-End Humanoid Soccer Control via Depth-Based Active Perception

Dec 2025–Present

*Yixiao Huo**, *Pengyuan Wang**, *Jiakang Jin**, *Yinan Han**, *et al.*

- Designed the perception-control architecture and led sim-to-real deployment for a 25-DOF humanoid (modified [Noetix E1](#)), integrating a 2-DOF active vision head to decouple perception from base locomotion.
- Developed a multimodal end-to-end RL framework centered on depth perception, and integrated auxiliary prediction heads within the vision encoder to robustly extract target latents and strengthen feature-driven state estimation.
- Architected adaptive task-switching policies for hierarchical dribbling and shooting, structuring searching and chasing as autonomous prerequisite stages for robust, vision-driven execution of target-oriented soccer maneuvers.

Autonomous Humanoid Exploration in Multi-Level Terrains

Jan 2026–Present

*Yinan Han**, *Yixiao Huo**, *Pengyuan Wang**, *Jiakang Jin**, *et al.*

- Orchestrated a goal-free exploratory framework for a 25-DOF humanoid (modified Noetix E1), using a 2-DOF active vision head to support proactive environmental scanning and heuristic pathfinding beyond fixed fields of view.
- Engineered a structural-feature encoder with an auxiliary terrain predictor to prioritize traversal toward high-variance topographies, enabling autonomous direction guidance based on localized depth complexity.
- Synthesized competitive reward dynamics that balance displacement efficiency against survival constraints, inducing emergent locomotion that autonomously modulates traversal speed according to real-time terrain risk.

COMPETITIONS

RoboMaster University Series: Embedded System Architecture

2023–2025

- Architected the embedded software framework and electrical topology for 4-swerve infantry and quadrotor platforms, and implemented a multi-layer state machine to ensure deterministic low-level motion control and system reliability.
- Developed a high-bandwidth communication protocol using bit-fields to maximize transmission frequency while maintaining data integrity, improving real-time throughput between the main controller and distributed actuators.
- Refined actuator dynamics through feed-forward cascaded PID and feedback control, achieving precise projectile velocity convergence and responsive gimbal tracking with zero overshoot for vision-coupled auto-aiming deployment.

ENGINEERING PROJECTS

General Motion Tracking on E1 Humanoid Platform2025

Internship at Noetix Robotics

- Reproduced GMT frameworks (BeyondMimic/HoloMotion) on the E1 platform, and optimized training via adaptive sampling and dataset cleaning, significantly improving tracking fidelity for complex, high-dynamic dance sequences.
- Engineered a Sim-to-Real pipeline in Isaac Lab by optimizing training architectures, and coordinated with hardware teams to deploy policies, achieving stable whole-body coordination and fluid imitation on physical robots.

Agile UAV Flight Control via Deep Reinforcement LearningJul–Sep 2025

Research project at FAST Lab, ZJU

- Constructed a high-mobility control stack integrating Jetson ORIN NX and PX4, and implemented cascaded PID controllers to decouple position and attitude, ensuring high-frequency response for agile flight maneuvers.
- Deployed a PPO-based RL framework for aggressive motion control, leveraging extensive domain randomization to bridge the sim-to-real gap and facilitate robust policy transfer from virtual environments to hardware.
- Accomplished autonomous execution of complex trajectories including high-speed figure-8 maneuvers, integrating the system via ROS to optimize data synchronization between high-level neural policies and low-level flight hardware.

SELECTED AWARDS

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| • First Prize in China University Robot Competition (RMUC) | (National Top 4) 2024
(National Top 8) 2025 |
| • First Prize in China University Robot Competition (RMUL) | (Regional 3rd Place) 2024
(Regional 2nd Place) 2025 |
| • First Prize in Zhejiang University Robot Competition | (2nd Place) 2024 |
| • Outstanding Student & Academic Excellence Award, ZJU | 2024, 2025 |
| • Zhejiang University Third-Class Scholarship (Top 20%) | 2024, 2025 |

SERVICE

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| • Team Leader ZJU RoboMaster Team “Hello World” | 2023–2025 |
| • Minister Zhejiang University Student Robotics Association (100+ members) | 2024–2025 |
| • Volunteer Opening Ceremony of Zhejiang University | 2023 |
| • Volunteer Zhejiang University Annual Sports Meet | 2025 |

SKILLS

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|-------------------|---|
| • Robot Platforms | Noetix E1, Noetix N2, DeepRobotics DR02 |
| • Programming | Python, PyTorch, C/C++ |
| • Software | IsaacGym/Lab, Gazebo, SolidWorks |
| • Hardware | Arduino/51/STM32, Circuit Design |
| • Others | ROS, Git, Linux, LaTeX |